



daham.serenedge.com

Daham Dissanayake

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SUMMARY

Third-year Computer Science undergraduate and Research Intern working on vision-based perception and reinforcement learning for robotics. Founder of SoterCare, an edge-first IoT and ML elderly monitoring startup, and of SerenEdge, an IT studio. Ships end-to-end systems across ML, embedded hardware, and infrastructure.

EXPERIENCE

Research Intern

June 2026 – Ongoing

Research Department, Informatics Institute of Technology (IIT)

Colombo, Sri Lanka

- RL research on the Unitree Go2 quadruped; administer the university GPU cluster via the Slurm-Deck gateway.

Founder & Technical Lead

2025 – Ongoing

SoterCare | Edge-first IoT and ML elderly monitoring startup

Sri Lanka

- Lead technical direction across firmware, on-device inference, backend, and mobile, from concept to a working edge-deployed prototype.

Founder

May 2026 – Ongoing

SerenEdge | serenedge.com

Sri Lanka

- IT studio delivering embedded firmware, ML pipelines, IoT fleets, and web platforms.

TECHNICAL SKILLS

ML & Computer Vision : PyTorch, TensorFlow, Isaac Lab, Reinforcement Learning, YOLOv11, OpenCV

Programming Languages : Python, C++, Java, TypeScript, JavaScript, SQL

Robotics & Embedded : Unitree Go2, Raspberry Pi, ESP32, ESP8266, Arduino, Edge Impulse, I2C, I2S

Infrastructure : Linux, SLURM, Docker, Git, Conda, PostgreSQL, Redis, SQLite, n8n

Full-Stack : Next.js, NestJS, FastAPI, React Native, Ballerina, Tailwind CSS

FEATURED PROJECTS

SoterCare | IoT & ML Elderly Care Startup | 2025 – Ongoing

<https://sotercare.com>

- Edge-first monitoring system that predicts falls and detects urinary incontinence in real-time for elderly residents.
- Wearable nodes stream to a Raspberry Pi gateway running Edge Impulse models on-device, keeping data local.
- **Tech**: Raspberry Pi, Edge Impulse, Python, C++, NestJS, React Native, PostgreSQL

MILO Robot | Free-Roaming Quadruped | 2026 – Ongoing

<https://github.com/DahamDissanayake/MILO-Robot>

- A quadruped robot that walks with an RL-trained neural network and remembers everyone it meets.
- A Raspberry Pi Zero 2W drives 8 servos with camera, mic, and IMU, offloading LLM and vision to a LAN GPU host.
- **Tech**: Python, Raspberry Pi Zero 2W, Reinforcement Learning, SQLite, WebSocket, LLM, I2C, Ollama, KG

Slurm-Deck | Secure SLURM GPU Job Gateway | 2026

<https://github.com/DahamDissanayake/slurm-deck>

- Secure terminal gateway for SLURM GPU job submission, running 24/7 for research users at IIT.
- Users land in a forced-command TUI over SSH; every action is logged to SQLite and JSONL before execution.
- **Tech**: Python, SLURM, SQLite, JSONL, TUI, Conda, PyTorch, SSH

AutoHub | Self-Hosted Automation OS | 2026

<https://github.com/DahamDissanayake/auto-hub>

- Self-hosted personal automation OS on a Raspberry Pi 5, being hardened ahead of an open-source release.
- One Docker Compose stack bundles a web terminal, plugin runner, Docker manager, n8n workflows, and Telegram alerts behind a Cloudflare Tunnel.
- **Tech**: Next.js, NestJS, Docker, n8n, PostgreSQL, Redis, Cloudflare Tunnel, WebSocket, JWT

OTHER PROJECTS

ImgHarvest | AI Image Dataset Harvester | 2026

🔗 <https://github.com/DahamDissanayake/img-harvest-ballerina>

- Microservices dataset harvester that refines search queries via Groq API and automates image collection for ML training; deployed on Choreo WSO2 under the GNU GPL.

VibeCheck | UI Debugging Chrome Extension | 2025

🔗 <https://github.com/DahamDissanayake/web-design-prompt-extension>

- Manifest V3 extension for overlaying precision grids and marking friction points, synthesizing structured AI prompts; live on the Chrome Web Store.

RelImage Agent | AI Community Avatars | 2026

🔗 <https://github.com/DahamDissanayake/RelImage-Agent>

- Generative avatar service giving communities a consistent visual identity, with a FastAPI backend handling generative workloads behind a Next.js frontend.

Dev-Pulse API | Real-time GitHub Metrics Aggregator | 2026

🔗 <https://github.com/DahamDissanayake/dev-pulse-api>

- Aggregates GitHub API data to generate a secure, real-time portfolio score via Ballerina and WSO2 Choreo.

Imposter Game | Custom Feature-Rich Alternative | 2025

🔗 <https://github.com/DahamDissanayake/Imposter-Game-But-Customized-V2>

- Built using React/Vite as a free alternative to paywalled apps; focuses on accessibility and seamless Vercel deployment.

View all projects at daham.serenedge.com | github.com/DahamDissanayake

EDUCATION & CERTIFICATIONS

BSc (Hons) in Computer Science (UG)

Informatics Institute of Technology (IIT) | University of Westminster, London

2025 – Ongoing

Cumulative Average: 81.6%

G.C.E. Advanced Level - Physical Science

Bandaranayake College, Gampaha | ICT - A, Physics, Combined Mathematics

Completed: Dec 2024

Sri Lanka

Professional Certificate in Machine Learning

Informatics Institute of Technology, Professional Development Unit (PDU)

Completed: 2025

Colombo

Professional Certificate in Marketing

Sri Lanka Institute of Marketing (SLIM)

Completed: 2023

Colombo

LEADERSHIP & VOLUNTEERING

Interact Club - Bandaranayake College

Executive Committee Member

2023 – 2024

Sri Lanka

IEEE CIS IIT (Prestige)

Event Volunteer

2025

Colombo

REFERENCES

Mr. Banu Athuraliya

Visiting Lecturer/SDGP Module Leader

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